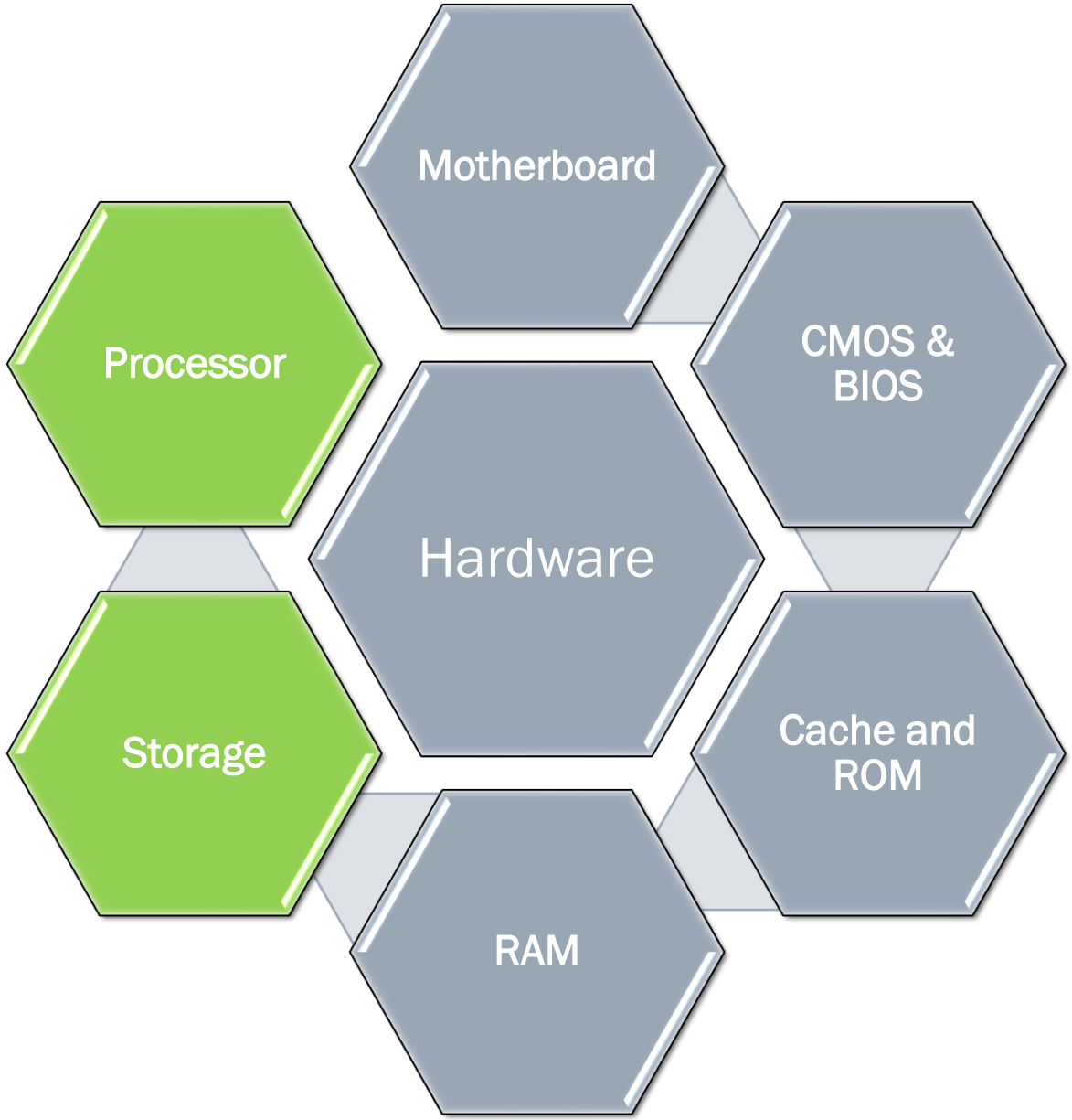
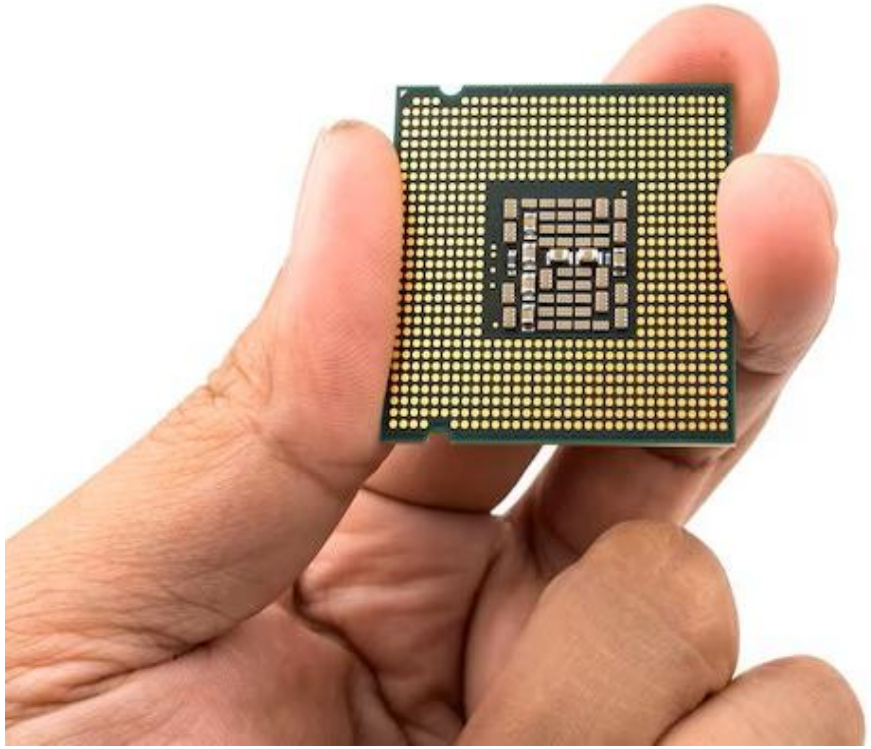
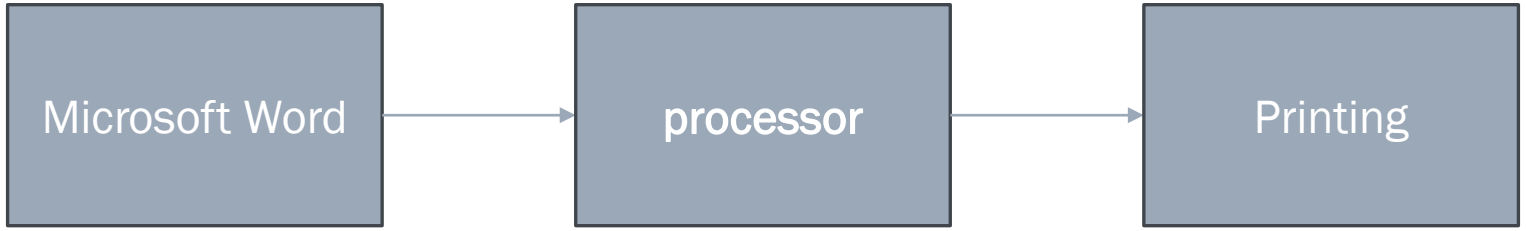

Motherboard
Processor
CMOS &
BIOS
Hardware
Storage
Cache and
ROM
RAM



1- The Processor

It’s the single most important component in any computer system — the **processor**, or **central processing unit (CPU)**, is the *heart* of the computer because it is responsible for telling all the other components what to do, according to the instructions given by the programs (software) running on the system.

For example, if you are using Microsoft Word and you want to print a document, Microsoft Word will tell the processor that the document needs to be printed. The processor will then communicate with the components responsible for printing the document. That's why we call it the **"processor"** — because it *processes* tasks. Every single task that happens on your computer is executed by the processor.



processor speed.

Now, when we talk about the processor, we must also mention a very important topic: **processor speed.** This is one of the most important factors in determining how fast your computer will be. Processor speed is usually expressed in **gigahertz (GHz)**, which indicates how many calculations the processor can make each second. Naturally, a higher number means more calculations and therefore a faster processor — for example, a **3.6 GHz** processor is faster than a **2.4 GHz** one.

As an example to better understand this point, let's look at the difference between an **Arduino microcontroller** and a **computer processor**.

Feature	Computer CPU	Arduino Microcontroller
Speed	GHz (billions/sec)	GHz (billions/sec)
Purpose	General computing	Hardware control
OS	OS	no
Multitasking	yes	no
Power Use	high	low
Example	Intel i7 @ 3.4 GHz	ATmega328P @ 16 MHz

Typical Speed Comparison

Device	Clock Speed	Relative Speed	Notes
Arduino Uno (ATmega328P)	16 MHz = 0.016 Ghz	1× (reference)	Basic 8-bit microcontroller
Modern Laptop/Desktop CPU (e.g., Intel i7, AMD Ryzen)	3.0–5.0 GHz	≈ 200–300× faster (per core)	64-bit, multi-core processor
With 8 cores (typical PC)	3.0–5.0 GHz × 8 cores	≈ 1,500–2,000× faster total	Parallel processing, cache, branch prediction, etc.

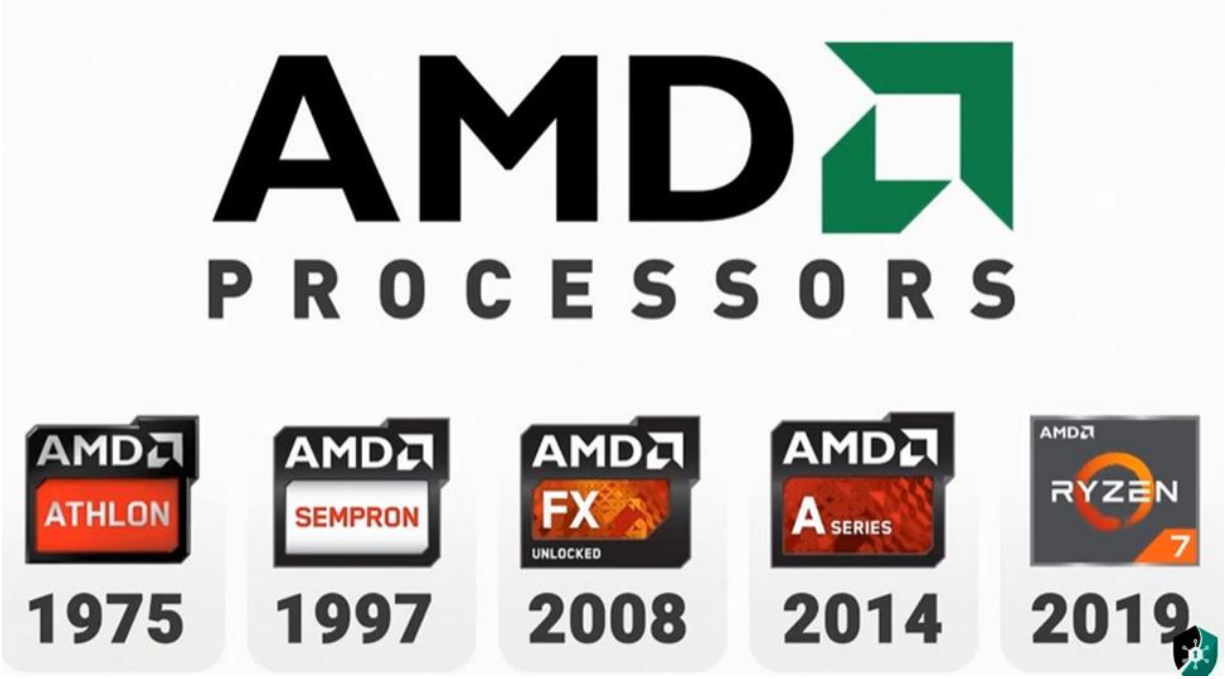
- **Clock Speed (GHz):** How many actions it can do per second.
- **Cores:** More cores = more tasks at the same time.

CPU	Cores	Speed	Use
i3	2	3.0 GHz	Basic work
i5	4	3.5 GHz	Office & design
i7	8	4.0 GHz	Gaming & heavy work

Processor Manufacturing Companies:

There are two major companies that manufacture computer processors — **AMD** and **Intel**.
There might be slight differences in their performance, as some processors require more cooling power. Keep in mind that processors generate heat while running, so they need a **cooling system**, which your laptop already has. Some processors require **more cooling** than others because they operate at **higher speeds**, which also makes them faster.

AMD


AMD
PROCESSORS
AMD ATHLON
1975
AMD SEMPRON
1997
AMD FX UNLOCKED
2008
AMD A SERIES
2014
AMD RYZEN 7
2019

intel

intel
CORE i7
vPro
9th Gen
intel
CORE i9
vPro
9th Gen
intel
CORE i5
vPro
9th Gen

How the CPU Works

The CPU always repeats three steps:

1. **Fetch** – The CPU gets an instruction from memory: “Add 2 and 3.”
2. **Decode** – Understand what to do , figures out what that means: “Use the add operation with numbers 2 and 3.”
3. **Execute** – Do the action , The CPU does the math and gets 5. It sends 5 back to memory/screen to show the answer.

💡 This is called the **Fetch–Decode–Execute Cycle,** This same 3-step cycle happens for everything: opening a file, typing a letter, playing a video—just different instructions.

Example:

When you click “Play” on a video:
CPU **fetches** the command,
decodes it (open video app),
executes it (plays the video).
This happens so fast — you don’t even notice!

Parts of the CPU

1. **CU** – The boss; tells other parts what to do
2. **ALU** – Does math and logic (add, compare, etc.)
3. **Registers** – Tiny, fast storage for quick data
4. **Cache** – Small, very fast memory for repeated tasks

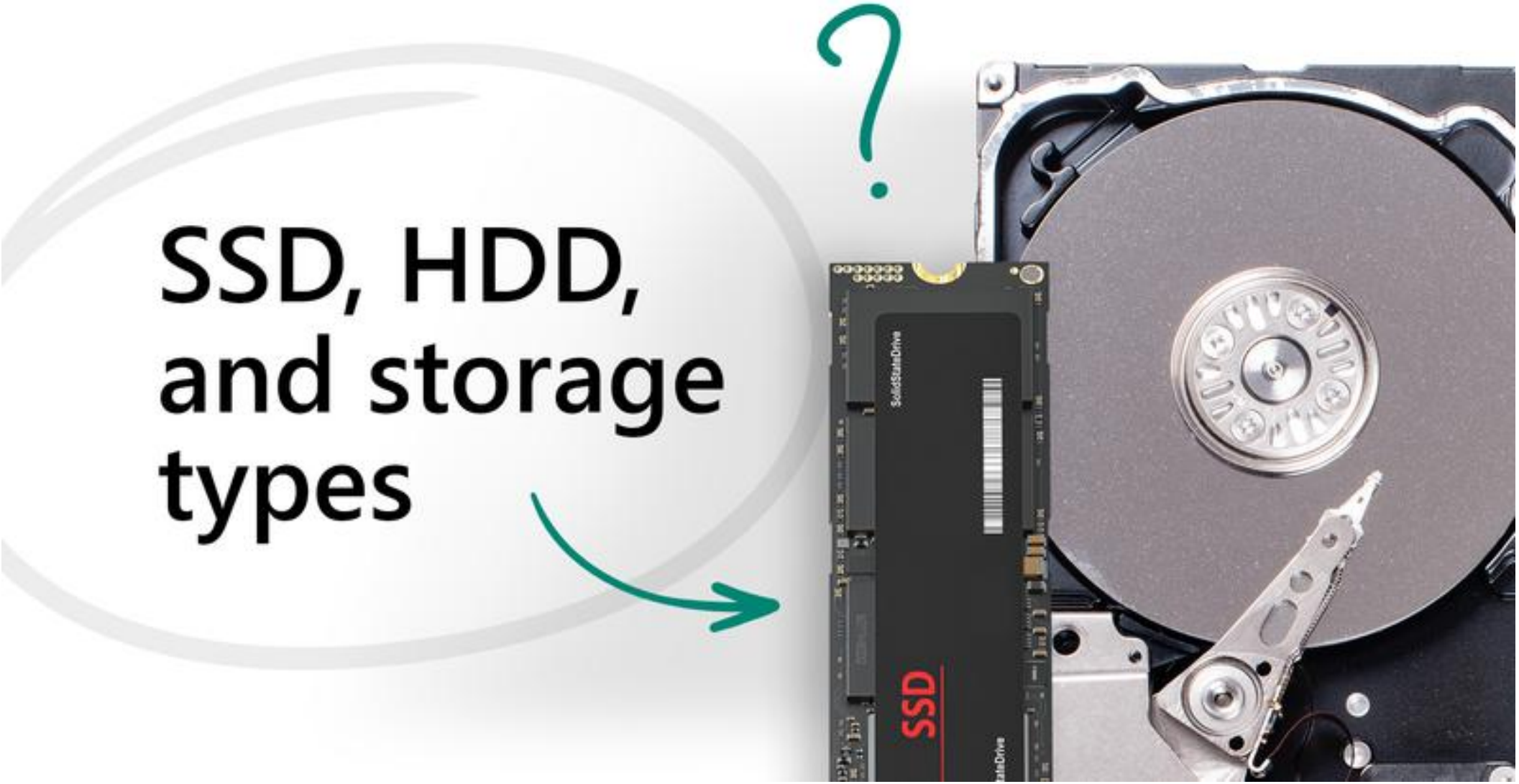
🧠 All of them work together to make the computer fast.

CPU Part	Like In Real Life	What It Does
CU	Manager	Gives orders
ALU	Calculator	Solves problems
Registers	Sticky notes	Keeps quick info
Cache	Desk drawer	Stores things you use a lot

Easy Example (CPU Parts working together)
Same "2 + 3" example, but who does what?
CU (Control Unit) = *The boss*
"Hey everyone, we have a job: add 2 and 3. ALU, get ready!"
Registers = *Tiny notepads*
They hold the numbers **2** and **3** for a moment.
ALU (Arithmetic Logic Unit) = *The calculator*
It adds 2 + 3 and gets **5**.
Cache = *Desk drawer of frequent stuff*
If we keep doing similar adds, the CPU keeps needed info close here so it's faster next time.

🧠 **All together:**
CU gives orders → Registers hold data → ALU calculates → Cache makes repeats faster → Result goes to memory/screen.

2- Storage



What Is Storage?
Storage is where the computer **keeps all the data** — both **short-term and long-term**.
It remembers your files, pictures, programs, and operating system even when the power is off.

Simple idea:
If the **CPU** is the brain, then **storage** is the memory or the bookshelf that keeps everything safe.

- **Types of Storage**

  1. Primary Storage (Inside the computer)
  Fast but small.
  Used while the computer is running.

  Examples:
  RAM (Random Access Memory) – short-term, temporary.
  Cache – very fast, helps the CPU work quicker.

📘 **Example:**
When you open a Word document, it loads into RAM so you can edit it quickly.

2. Secondary Storage (Long-term storage)
Keeps data even when the computer is turned off.
Examples:
Hard Disk Drive (HDD) – stores large files; cheaper but slower.
Solid State Drive (SSD) – faster, newer, no moving parts.
📘 **Example:**
Your photos, music, and apps are saved on the SSD/HDD.

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3. Portable or External Storage

Easy to carry and share data between devices.

Examples:

USB flash drives

External hard drives

Memory cards

📘 **Example:**

You save your presentation on a USB and plug it into another computer.

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4. Cloud Storage
Saves your files on the internet instead of your computer.
You can access them from anywhere.
Examples:
- Google Drive
- Dropbox
- iCloud

📘 **Example:**
You upload a project to Google Drive so your teacher can open it from another city.

Type	Speed	Keeps data when off?	Example
RAM	Very Fast	❌ No	Running apps
SSD	Fast	✅ Yes	System files
HDD	Medium	✅ Yes	Photos, videos
USB	Slow–Medium	✅ Yes	Backup files
Cloud	Depends on internet	✅ Yes	Online sharing